

Sina Sajadmanesh

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SUMMARY

Senior AI Engineer specializing in multimodal foundation models and video understanding, with a track record of leading end-to-end R&D and translating research into efficient, production-ready ML systems across heterogeneous hardware.

EDUCATION

Swiss Federal Institute of Technology (EPFL) , PhD in Electrical Engineering	Aug 2023
Sharif University of Technology , MSc in Information Technology Engineering	Sept 2016
University of Isfahan , BSc in Computer Software Engineering	Feb 2014

EXPERIENCE

Sony AI , Senior AI Engineer – Zurich, Switzerland	Jan 2026 – present
- Leading end-to-end R&D of a multimodal video understanding system using natural-language instructions to flexibly segment, retrieve, and annotate long-form video - Deployed video-language action recognition models on Apple Silicon for Sony's Remote Camera Tool, building an end-to-end Swift/Core ML inference pipeline on the Neural Engine with up to $53\times$ faster inference and $19\times$ lower peak memory - Built agentic AI workflows for automated model optimization, improving inference speed and memory footprint of LLMs and VLMs for Apple Silicon - Architected a modular ML acceleration and deployment framework supporting diverse hardware targets (Nvidia GPUs, Qualcomm NPUs, Apple Silicon) - Developed compact vision-language models supporting image/video understanding and localization tasks, achieving competitive performance with reduced computational requirements	
Sony AI , AI Engineer – Zurich, Switzerland	Oct 2023 – Dec 2025
- Designed a modular and flexible geometric optimization algorithm, enabling NeurIPS'25 Spotlight paper - Led optimization of Argus VFM inference stack across heterogeneous hardware, achieving up to $61\times$ speedup across 17 vision tasks - Led development and integration of 7 vision tasks, including depth estimation, surface normal prediction, super-resolution, and denoising, into a unified 17-task VFM - Architected and led development of a modular multi-task learning framework unifying OpenMMLab libraries into a single distributed training platform powering Sony's Argus VFM	
Idiap Research Institute , Research Assistant – Martigny, Switzerland	May 2019 – Aug 2023
- Worked on differentially private machine learning with graph neural networks 3 papers in top-tier conferences (CCS, USENIX Security, and WSDM) with 340+ citations 7 invited talks at top universities and research institutions, including Imperial College, UIC, and Twitter 6 open-source projects with 130+ stars on GitHub 1 short course taught on "Trustworthy Machine Learning" at Artificial Intelligence Doctoral Academy - Finalist in CSAW Applied Research Competition for the best paper award in computer security in Europe	
Brave Software , Research Intern – Remote	Mar 2022 – May 2022
Developed a novel privacy-preserving federated learning framework for neural bandit models under client heterogeneity using PyTorch, Flower, Dask, and Tensorflow-Lite	

SKILLS AND EXPERTISE

Foundation & Generative Models: Vision & Multimodal Models (ViT, CLIP, LLaVA) · Large Language Models (LLaMA, GPT, Qwen) · Parameter-Efficient Fine-Tuning (LoRA) · Self-Supervised Learning (MAE, DINO)

Model Optimization & Deployment: TensorRT · ONNX · Qualcomm QNN / SNPE / QAIRT · Apple CoreML / MLX · Optimum · Quantization · Graph Optimization · Performance Profiling

Machine Learning Frameworks: PyTorch · Hugging Face (Transformers, PEFT, Accelerate, Datasets) · OpenMMLab (MMCV, MMEEngine, MMDetection)

Agentic AI & LLM Systems: LangChain · LangGraph · MCP · Tool Calling · RAG · Multi-Agent Systems · Workflow Orchestration

Engineering & Tooling: Python · C++ · Bash · SQL · Data Processing (NumPy, Pandas, torchvision) · Environment & Packaging (uv, ruff) · CI/CD (Docker, GitHub Actions)

COMMUNITY AND PROFESSIONAL SERVICE

Invited Speaker: Imperial College London (2023, 2020), University of Illinois at Chicago (2022), L3S Research Center (2022), GNN User Group Meetup (2021), Twitter ML Seminar (2021)

Organizing / Program Committee: Privacy and Fairness in AI for Health (2023), PPAI (2024), WiseML (2023), PAIR2Struct (2022), DPML (2021)

Reviewer: NeurIPS (2026, 2024), CVPR (2026), TIFS (2025), IMWUT (2024), TDSC (2023), LoG (2023, 2022), AISTATS (2023), AIJ (2022), TBD (2021), TIST (2020), SNAM (2020), WWWJ (2018)

SELECTED PUBLICATIONS

StellA: Subspace Learning in Low-rank Adaptation using Stiefel Manifold Zhizhong Li, Sina Sajadmanesh, Jingtao Li, Lingjuan Lyu NeurIPS 2025 🌟 Spotlight (top 3%)	Dec 2025
Argus: A Compact and Versatile Foundation Model for Vision Weiming Zhuang, Chen Chen, Zhizhong Li, Sina Sajadmanesh, et al. CVPR 2025	June 2025
ProGAP: Progressive Graph Neural Networks with Differential Privacy Guarantees Sina Sajadmanesh, Daniel Gatica-Perez WSDM 2024	Mar 2024
GAP: Differentially Private Graph Neural Networks with Aggregation Perturbation Sina Sajadmanesh, Ali Shahin Shamsabadi, Aurélien Bellet, Daniel Gatica-Perez USENIX Security 2023	Aug 2023
Locally Private Graph Neural Networks Sina Sajadmanesh, Daniel Gatica-Perez CCS 2021	Nov 2021
A Hybrid Deep Learning Architecture for Privacy-Preserving Mobile Analytics Seyed Ali Osia, Ali Shahin Shamsabadi, Sina Sajadmanesh, et al. IEEE Internet of Things Journal	May 2020
Continuous-Time Relationship Prediction in Dynamic Heterogeneous Information Networks Sina Sajadmanesh, Sogol Bazargani, et al.	Aug 2019

TKDD

Kissing Cuisines: Exploring Worldwide Culinary Habits on the Web

Sina Sajadmanesh, Sina Jafarzadeh, Seyed Ali Ossia, et al.

WWW 2017

Apr 2017